IMO Intensive Training 2019
Symmedian
8th June, 2019

1. In triangle ABC, the reflection of the A-median with respect to the internal angle bisector of $\angle A$ is called the ***A*-symmedian**. Let ω be the circumcircle of triangle ABC.
   (a) Let the A-symmedian meet BC at D. Prove that $\dfrac{BD}{DC} = \left(\dfrac{BA}{AC}\right)^2$.
   (b) If $\angle A \neq 90^\circ$, denote by T the intersection of tangents to ω at points B and C. Prove that line AT is the A-symmedian in triangle ABC.
   (c) Let the A-symmedian meet ω for the second time at S. Prove that $BS \cdot AC = CS \cdot AB$.
2. In triangle ABC, suppose the A-symmedian is perpendicular to BC. Prove that triangle ABC is either isosceles or right-angled.
3. Suppose A, B, C, D all lie on circle ω, where neither AC nor BD is a diameter of ω. Denote by P the intersection of tangents to ω at points A and C, and Q the intersection of tangents to ω at points B and D. Show that A, C, Q are collinear if and only if B, D, P are collinear.
4. Let ABC be an isosceles triangle with base BC. Let P be a point inside the triangle ABC such that $\angle CBP = \angle ACP$. Denote by M the midpoint of the base BC. Show that $\angle BPM + \angle CPA = 180^\circ$.
5. Let ABC be a scalene triangle. The internal angle bisector of $\angle A$ intersects the side BC at D and the circumcircle Ω of triangle ABC again at E. Circle ω with diameter DE cuts Ω again at F. Prove that AF is the symmedian of triangle ABC.
6. Let ABC be a triangle. Suppose K and L are points on AB and AC respectively. Let P be the second intersection of the circumcircles of triangles ABL and AKC. Show that AP is the A-symmedian of triangle ABC if and only if $KL // BC$.
7. In acute triangle ABC, let D, E, and F be the feet of the altitudes from A, B, and C respectively, and let L, M, and N be the midpoints of BC, CA and AB, respectively. Lines DE and NL intersect at X, lines DF and LM intersect at Y, and lines XY and BC intersect at Z. Find $\dfrac{ZB}{ZC}$ in terms of AB, AC, and BC.
8. Let ABC be a triangle. Denote by H the orthocentre of triangle ABC and M the midpoint of BC. Suppose K is a point on AM such that $HK \perp AM$. Let D be the reflection of K with respect to line BC. Prove that AD is the A-symmedian of triangle ABC.
9. Let ABC be an acute, scalene triangle, and let M, N and P be the midpoints of BC, CA and AB, respectively. Let the perpendicular bisectors of AB and AC intersect ray AM in points D and E, respectively, and let lines BD and CE intersect at point F, inside triangle ABC. Prove that A, N, F, P are concyclic.
10. Let AD be the inner angle bisector of the triangle ABC. The perpendicular on the side BC at the point D intersects the outer bisector of $\angle CAB$ at point I. The circle with center I and radius ID intersects the sides AB and AC at points F and E respectively. A-symmedian of ΔAFE intersects the circumcircle of ΔAFE again at point X. Prove that the circumcircles of ΔAFE and ΔBXC are tangent.